1 **(a)** $\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right)$ **[M1]**

$(1, 1)$ **[A1]**

(b) $(4-(-2))^2 + (-3-5)^2$ or $6^2 + (-8)^2$ **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

2 **(a)** $\left(\frac{1+7}{2}, \frac{4+(-4)}{2}\right)$ **[M1]**

$(4, 0)$ **[A1]**

(b) $(7-1)^2 + (-4-4)^2$ oe **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

3 **(a)** Uses the midpoint formula: $\left(\frac{2+8}{2}, \frac{-3+5}{2}\right)$ **[M1]**

$(5, 1)$ **[A1]**

(b) Uses the distance formula: $\sqrt{(8-2)^2 + (5-(-3))^2}$ **[M1]**

$\sqrt{6^2+8^2}$ or $\sqrt{36+64}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

4 **(a)** $A = (-2, 1)$ **[B1]**

$B = (4, 9)$ **[B1]**

(b) $\left(\frac{-2+4}{2}, \frac{1+9}{2}\right)$ **[M1]**

$(1, 5)$ **[A1]**

(c) $\sqrt{(4-(-2))^2 + (9-1)^2}$ or $\sqrt{6^2+8^2}$ **[M1]**

10 **[A1]**

[Total Marks: 6]

5 **(a)** Calculates length OP: $\sqrt{(-3)^2+4^2}$ or $\sqrt{9+16}$ **[M1]**

5 **[A1]**

(b) Recognises $O(0, 0)$ is the midpoint of PQ, so sets up $\frac{-3+x}{2} = 0$ and $\frac{4+y}{2} = 0$ OR uses

translation from P to O **[M1]**

$(3, -4)$ **[A1]**

(c) Sets up equation for length OR: $k^2 + (\sqrt{11})^2 = 5^2$ **[M1]**

$k^2 + 11 = 25$

$k^2 = 14$ [M1]

$-\sqrt{14}$ **[A1]**

[Total Marks: 7]

6 (a) $\frac{-2+x}{2} = 3$ or $\frac{4+y}{2} = -1$ **[M1]**

$(8, -6)$ **[A1]**

(b) $PQ = \sqrt{(8-(-2))^2 + (-6-4)^2}$ **[M1]**

$\sqrt{100+100}$ or $\sqrt{200}$ **[M1]**

$10\sqrt{2}$ **[A1]**

(c) Let $R = (0, y)$ and form $(0-(-2))^2 + (y-4)^2 = (0-8)^2 + (y-(-6))^2$ **[M1]**

$4 + y^2 - 8y + 16 = 64 + y^2 + 12y + 36$ **[M1]**

$-20y = 80$ giving $y = -4$ **[A1]**

$(0, -4)$ **[A1]**

[Total Marks: 9]

7 (a) Finds coordinates of the midpoint in terms of p: $\left(\frac{4p}{2}, \frac{3p-1}{2}\right)$ or $\left(2p, \frac{3p-1}{2}\right)$ **[M1]**

Substitutes into the equation of the line: $\frac{3p-1}{2} = 2p + 1$ **[M1]**

-3 **[A1]**

(b) Sets up the distance squared equation: $(3p-p)^2 + (p-2-(2p+1))^2 = 6^2$ **[M1]**

Simplifies to $(2p)^2 + (-p-3)^2 = 36$ **[M1]**

Expands to $4p^2 + p^2 + 6p + 9 = 36$ and rearranges to $5p^2 + 6p - 27 = 0$ with no errors seen **[A1]**

(c) Factorises to $(5p-9)(p+3) = 0$ or uses correct quadratic formula **[M2]**

Award [M1] for a partial factorisation or one error in the quadratic formula

1.8 (or $\frac{9}{5}$) and -3 **[A1]**

[Total Marks: 9]

8 (a) $\left(\frac{-1+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(2, 5)$ **[A1]**

(b) $(2-(-1))^2+(5-2)^2$ or $\frac{1}{2}\sqrt{(5-(-1))^2+(8-2)^2}$ oe **[M1]**

$\sqrt{9+9}$ or $\sqrt{18}$ **[M1]**

$3\sqrt{2}$ **[A1]**

(c) $\frac{2+x}{2}=\frac{5}{2}$ and $\frac{5+y}{2}=\frac{3}{2}$ oe **[M1]**

$x=3$ and $y=-2$ to give $D(3,-2)$ **[A1]**

Length $AD=\sqrt{(3-(-1))^2+(-2-2)^2}$ oe **[M1]**

$\sqrt{16+16}$ or $\sqrt{32}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Total Marks: 10]

9 (a) Uses distance formula: $\sqrt{(4-(-2))^2+(-3-5)^2}$ **[M1]**

Simplifies to $\sqrt{6^2+(-8)^2}$ or $\sqrt{36+64}$ **[M1]**

10 **[A1]**

(b) Uses midpoint formula: $\left(\frac{-2+4}{2},\frac{5+(-3)}{2}\right)$ **[M1]**

$(1,1)$ **[A1]**

[Total Marks: 5]

10 (a) $(7-3)^2+(5-k)^2=5^2$ or equivalent **[M1]**

$16+(5-k)^2=25$ or $(5-k)^2=9$ **[M1]**

$k=2$ and $k=8$ **[A1]**

(b) Identifies $P(3,2)$ to use in midpoint calculation: $\left(\frac{3+7}{2},\frac{2+5}{2}\right)$ **[M1]**

$(5,3.5)$ **[A1]**

[Total Marks: 5]

11 (a) Sets up an equation for at least one coordinate, e.g., $\frac{-3+x}{2}=1$ or $\frac{5+y}{2}=-2$ **[M1]**

$(5,-9)$ **[A1]**

(b) Uses distance formula with $P(-3,5)$ and their $Q(5,-9)$: $\sqrt{(5-(-3))^2+(-9-5)^2}$ **[M1]**

$\sqrt{8^2+(-14)^2}$ or $\sqrt{64+196}$ or $\sqrt{260}$ **[M1]**

16.1 **[A1]**

[Total Marks: 5]

12 (a) $(k-(-3))^2+(-1-5)^2=10^2$ oe **[M1]**

$(k+3)^2+36=100$ or $k^2+6k+9+36=100$ **[M1]**

$(k+3)^2 = 64$ or $k^2 + 6k - 55 = 0$ **[M1]**

$k = 5$ or $k = -11$ **[A1]**

(b) Chooses $k = 5$ to establish coordinates of Q as $(5, -1)$ **[M1]**

$(1, 2)$ **[A1]**

[Total Marks: 6]

13 (a) $\left(\frac{-3+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(1, 5)$ **[A1]**

(b) Radius $r = \sqrt{(1-(-3))^2 + (5-2)^2}$ or $\sqrt{4^2+3^2} = 5$ **[M1]**

Area $= \pi \times 5^2$ **[M1]**

78.5 **[A1]**

(c) Distance $CD = 5 \implies (k-1)^2 + (8-5)^2 = 5^2$ **[M1]**

$(k-1)^2 + 9 = 25 \implies (k-1)^2 = 16$ **[M1]**

$k = 5$ and $k = -3$ **[A1]**

[Total Marks: 8]

14 (a) Sets up the distance formula: $\sqrt{(3p-2p)^2 + (4p-p)^2} = 5\sqrt{10}$ **[M1]**

Simplifies the terms inside the root: $\sqrt{p^2 + (3p)^2} = 5\sqrt{10}$ **[M1]**

Squares both sides and simplifies: $10p^2 = 250$ **[M1]**

$p = 5$ (rejects -5 as $p > 0$) **[A1]**

(b) Evaluates coordinates of $A(10, 5)$ and $B(15, 20)$ **[B1]**

Sets up a valid expression for the coordinates of C, e.g., x-coordinate $= 10 + \frac{2}{5}(15-10)$ or y-coordinate $= 5 + \frac{2}{5}(20-5)$ **[M1]**

$(12, 11)$ **[A1]**

(c) Uses the midpoint formula, e.g. $\frac{15+x}{2} = 12$ and $\frac{20+y}{2} = 11$, or $D = 2C - B$ **[M1]**

$(9, 2)$ **[A1]**

[Total Marks: 9]

15 (a) $AB = \sqrt{(5-(-3))^2 + (6-2)^2} = \sqrt{80}$ $(= 8.94\ldots)$ **[M1]**

$BC = \sqrt{(9-5)^2 + (-2-6)^2} = \sqrt{80}$ $(= 8.94\ldots)$ **[M1]**

$AC = \sqrt{(9-(-3))^2 + (-2-2)^2} = \sqrt{160}$ $(= 12.6\ldots)$ **[M1]**

$AB = BC$ (and $ACeAB$), therefore triangle ABC is isosceles **[A1]**

(b) (i) $\left(\frac{-3+9}{2}, \frac{2+(-2)}{2}\right)$ **[M1]**

$(3, 0)$ **[A1]**

(ii) $BM = \sqrt{(5-3)^2 + (6-0)^2}$ $(= \sqrt{40})$ **[M1]**

6.32 **[A1]**

(c) $\frac{1}{2} \times \sqrt{160} \times \sqrt{40}$ or $\frac{1}{2} \times \sqrt{80} \times \sqrt{80}$ **[M1]**

40 **[A1]**

(d) $\frac{5+x}{2} = 3$ and $\frac{6+y}{2} = 0$ (or vector translation $B + \overrightarrow{AD}$) **[M1]**

$(1, -6)$ **[A1]**

[Total Marks: 12]